HW01

Intro1:

To act rationally is to choose the action that is expected to best achieve one's goals, given the available information, possible actions, and consequences. Rational action maximizes expected utility given the available information I:

$$a^* \in \arg\max_a \mathbb{E}[U|I,a]$$

Rationality concerns the decision method; selfishness concerns the goal.

$$U = U_{self}\ (selfish),\quad U = \alpha U_{self} + \beta U_{others}\ (\not\in)$$

Example: compare price and quality, then choose the best expected value. A rational choice may still have a bad outcome under uncertainty

Intro2:

Example: ~~esh~~ exhaustive game-tree search under a short time limit. For branching factor b and search depth d,

$$N_{leaves} = b^d,\quad N_{nodes} = \sum_{k=0}^{d} b^k = \frac{b^{d+1}-1}{b-1}\ (b>1)$$

Humans have limited memory and calculation speed. Computers can systematically enumerate and evaluate many moves; AI need not imitate human thinking.

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Intro 3

LLM: Codex; original discussion: October 8, 2026.

Prompt: Why did multilayer learning take yeors despite XOR needing only one extra hidden layer?

x_1	x_2	z_1(OR)	z_2(AND)	y(XOR)
0	0	0	0	0
0	1	1	0	1
1	0	1	0	1
1	1	1	1	0

For a single linear threshold, XOR would require

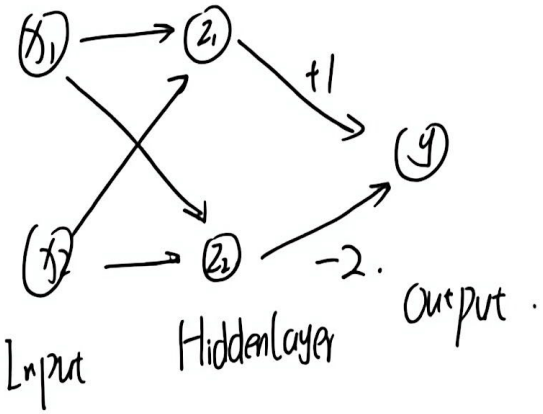
$$b<0,\ w_1+b\geq 0,\ w_2+b\geq 0,\ w_1+w_2+b<0.$$

But. $w_1+w_2+b \geq -b>0$: contradiction. Thus XOR is not. linearly separable.

Let $H(t) = \mathbb{1}[t>0]$. A nonlinear hidden layer solves XOR:

$$z_1 = H(x_1+x_2-0.5), \qquad z_2 = H(x_1+x_2-1.5),$$

$$y = H(z_1-2z_2-0.5)$$



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Difficulty: learning hidden weights, not drawing another layer.

For differentiable activations, the chain rule gives credit assignment:

$$\frac{\partial L}{\partial w^{(1)}} = \frac{\partial L}{\partial y}\frac{\partial y}{\partial z}\frac{\partial z}{\partial w^{(1)}}, \quad w \leftarrow w - \eta \nabla_w L.$$

Back-~~progo~~ propagation + nonlinear differentiable units + gradient optimization make multilayer learning practical. The hard-threshold network above demonstrates representation, not differentiable training.

Year	Milestone
1969	Minsky–Papert: Perceptrons [1]
1974	Werbos: backward differentiation for learning [3]
1986	Rumelhart–Hinton–Williams: back-propagation milestone [2]

$1986 - 1970 = 16$ years

The 1986 work popularized multilayer learning; it was not the first invention of hidden layers. "Dead around 1970" is a simplified description of declining interest.

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